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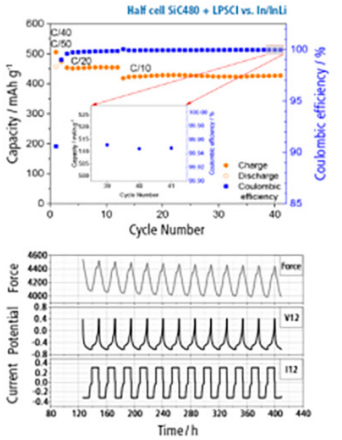
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A Novel Voltammetric Strategy of Furazidin Determination with the Use of Co-Ferrierite/Mesoporous Carbon Modified Glassy Carbon Electrode

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Furazidin (FUR) is a 5-nitrofuran derivative with a well-known and unique antibacterial activity, commonly applied in the treatment of urinary tract infections. FUR has been used regionally and globally in medicine, resulting in its continuous release into the natural ecosystem. Therefore, an accurate determination of trace levels of FUR in complex samples is highly reasonable and desirable. In this context, we propose a simple, fast, and highly sensitive electroanalytical method of FUR determination by means of differential pulse voltammetry with the use of Co-ferrierite/mesoporous carbon modified glassy carbon electrode (CoFER/MC-GCE). The advantageous features and electrochemical properties of CoFER/MC-GCE were confirmed by morphological and textural research, including X-ray fluorescence, Fourier-transform infrared spectroscopy, nitrogen adsorption/desorption method, and scanning electron microscopy coupled with energy dispersion spectroscopy measurements, as well as electrochemical studies involving cyclic voltammetry and electrochemical impedance spectroscopy (EIS). The linear response of CoFER/MC-GCE was achieved in the FUR concentration ranges of $0.02 - 0.29$ mg l^{-1} and $0.33 - 0.48$ mg l^{-1} with a detection limit of 3.9 μg l^{-1}. The acceptable results of pharmaceuticals, wastewater, and biological fluids analysis confirmed the analytical usefulness of the proposed voltammetric strategy of FUR determination using CoFER/MC-GCE.
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Furazidin (FUR), also known by the generic name furagin, is a 5-nitrofuran derivative with a well-known and unique antibacterial activity.[1] Chemically, furazidin (1-[(E)-[(E)-3-(5-nitrofuran-2-yl) prop-2-enylidene]amino]imidazolidine-2,4-dione), CAS number 1672–88–4, belongs to the group of hydantoins, i.e., heterocyclic compounds containing an imidazolidine substituted by a ketone group at positions 2 and 4. Through a wide spectrum of activity against both Gram-positive and Gram-negative microorganisms, FUR is applied in the treatment of urinary tract infections (UTIs), cystitis, and bacteriuria.[2,3] According to the literature data, urinary tract diseases are among the most common forms of bacterial infection worldwide.[4–6] It is estimated that about 50% of women and 10% of men experience an episode of UTI in their lifetime. Among the pediatric population, UTIs affect about one-tenth of the children with a recurrence rate between 30 and 50%. In view of the above, guidelines of European scientific societies recommend nitrofuran derivatives as the first-line treatment of uncomplicated UTIs.[7]

Currently, furazidin is commonly used mainly in the states of the former Soviet Union and some Eastern European countries.[8] Among these countries, Poland is the only one in the European Union where FUR is available as an over-the-counter (OTC) drug. Widespread access to furazidin-containing medications in Poland results in limited physician visits and more frequent self-treatments, which in many cases leads to their too hasty or even inappropriate use without a medical recommendation. This, in turn, leads to easy overdose and the possibility of various interactions with other medications taken. Although furazidin is a safe and effective antibacterial agent, the occurrence of anaphylactic shock or lung reaction associated with its administration has been reported.[9,10] On the other hand, the high use of FUR is one of the factors related to increased antimicrobial resistance, among which the resistance rate to nitrofuran derivatives exceeds 30% in Poland, while in other European countries it did not exceed 3%.[7,11] Finally, the high consumption of furazidin combined with low absorption by the human organism has been accompanied by its continuous discharge into the natural ecosystem. As a result of extraction from the body primarily through urine, a substantial amount of FUR can be released into wastewater, from which it may reach surface waters, and groundwaters, and, potentially, drinking water. According to the scientific suggestion,[12] furazidin, similar to other antibacterial pharmaceutical forms of the nitrofurans group, possesses the ability to interfere with living organisms at very low concentrations.[13] However, knowledge about the possible ecological impact and human toxicity of FUR via environmental exposure routes is still limited, and its presence is not routinely monitored by any national environmental legislation in Poland.

In view of the above, the determination of FUR in complex environmental samples is fundamental to understanding its behavior in aquatic ecosystems and thus the assessment of the risk of hazard to the environment and public health. Since the most commonly reported pharmaceuticals [12,14,15] have been determined at the concentrations ranging from ng l^{-1} to mg l^{-1} depending on the matrices of the aqueous environment, the development of analytical methods for the trace analysis of furazidin is highly reasonable and desirable. The literature reports related to the quantification of FUR are very limited. Among a few available papers, the application of capillary thin-layer chromatography,[16] amperometric biosensor,[17] UV–vis spectrophotometry,[18] and ion-selective electrode[19] has only been published. At the same time, the literature data does not provide any reports concerning the determination of FUR with the use of the voltammetric methods. The indisputable advantages of voltammetric approaches, in comparison to other analytical methods, are a simple procedure of sample preparation, a short time of a single analysis, excellent sensitivity and accuracy, a low detection limit, and high reproducibility. These characteristics have resulted in the development of numerous voltammetric sensors dedicated to the determination of different kinds of pharmaceuticals, not only in the environmental, but also in the biological, medication, and food matrices.[20–34]

Significant improvement in the sensitivity, stability, and selectivity of electrochemical sensors in the presence of common interfering species can be achieved by the modification of the working electrode with the application of various advanced nanomaterials and functionalized nanostructures.[35] One of them are zeolites, i.e., hydrated, micro- and mesoporous, three-dimensional crystalline inorganic

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solids, which belong to the family of tectoalauminosilicates. The principle advantages of zeolites include high surface area and microporosity, resistance to extreme thermal and chemical conditions, insolubility in most organic solvents and acid media, as well as availability in nature connected with low cost of exploration.[36,37] In the midst of the exceptional features of zeolites as electrode modifiers, the molecular sieving properties and high cation exchange capacity (CEC) play a crucial role. The discrimination/recognition of shape, size, and even charge displayed by zeolites enables selective accumulation of analyte on the surface of the zeolite-modified electrode (ZME).[38] In turn, the high CEC, depending on the nature, number, and availability of the cation exchange sites in zeolites structure, creates the possibility to receive their exchanged form with enhanced catalytic activity.[39] These unique electrocatalytic properties can be utilized in electrochemical analysis, provided that the receptor layer formed on the electrode surface with the presence of zeolite forms a nanostructural composite, which must be additionally doped with a material that ensures good electrical conductivity. Zeolites are insulators, therefore, as a conductive material uniformly dispersed in a polymeric matrix, various carbon-based, metallic, or oxide nanomaterials are used. As a result of their high chemical stability, various carbon materials, like graphite,[40] carbon black,[41] and graphene oxide,[42] are reported for this purpose. The obtained synergistic electrochemical effects resulting from the combination of zeolite with carbon nanomaterial allow to develop ZME suitable for the analysis of both inorganic and organic species.[43–46]

This paper introduces a simple and fast voltammetric protocol for furazidin determination in samples of pharmaceutical, biological, and environmental origin with the use of a novel sensor based on a glassy carbon electrode (GCE) modified with Co-ferrierite and mesoporous carbon nanocomposite (CoFER/MC-GCE). This approach is presented for the first time. As shown by the results of cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) measurements, the electrochemical performance of the developed sensor, fabricated by a low-cost and uncomplicated drop-casting method, is influenced by particular components of the modifying layer. Simultaneously, the reaction mechanism of FUR reduction on the surface of CoFER/MC-GCE was proposed based on the CV and differential pulse voltammetry (DPV) studies. The determination of furazidin in commercially available pharmaceutical formulations and certified reference materials (CRMs) of human serum, urine, and wastewater confirmed the analytical usefulness of the presented voltammetric strategy.

Experimental

Chemicals and reagents.—Nanopowder of mesoporous carbon (graphitized carbon) was received from Sigma-Aldrich (product number 699624) and used without any pretreatment. Synthetic zeolite powder of NH_4-ferrierite (CP 914 C) was purchased from Zeolyst International (Pennsylvania, United States). Dichloromethane, polystyrene (average M_w∼35,000), and acetone solution suitable for HPLC (⩾ 99.9%) were obtained from Honeywell Inc., Sigma-Aldrich, and Avantor Performance Materials Poland S.A., respectively.

All reagents were of analytical grade and used as received. The double-distilled water was used to prepare all the aqueous solutions. A fresh, standard stock solution of furazidin at the concentration of 1000 mg l^{-1} was prepared weekly by dissolving an appropriate amount of reagent (API Reference Standard, TRC, Canada) in water, protected from the light, and stored at 4° C in the fridge. Working solutions of FUR with lower concentrations were obtained just before measurement through dilution of the standard stock solution with double-distilled water. 0.1 mol l^{-1} phosphate buffer solution with pH between 6 and 7.4 was prepared by mixing an appropriate volume of 0.2 mol l^{-1} NaH_2PO_4 with 0.2 mol l^{-1} Na_2HPO_4 (Avantor Performance Materials Poland S.A.). Other chemicals, i.e., KCl, Na_3PO_4, $NaNO_3$, $MgSO_4{\cdot}7H_2O$, $CaCl_2{\cdot}6H_2O$, KNO_3, $Co(NO_3)_2{\cdot}6H_2O$, cetrimonium bromide (CTAB), and potassium hexacyanoferrate(II) and (III), were obtained from Avantor Performance Materials Poland S.A. AAS standard solutions of Cu (II), Cd(II), and Pb(II) (1000 mg l^{-1}), as well as Triton X - 100, sodium dodecyl sulfate (SDS), humic and uric acid, titanium dioxide, starch, cellulose, and magnesium stearate were purchased from Merck and Sigma-Aldrich, respectively. Certified Reference Materials (CRMs) of Waste Water (SPS-WW1 Batch 114 and SPS-WW2 Batch 113) were received from Spectrapure Standards AS, Oslo, Norway. The synthetic human plasma and urine were obtained from Biowest (France) and Medichem Diagnostica GmbH & Co. (Germany), respectively.

Instrumentation.— *Modifiers characterization.*—X-ray fluorescence measurement (XRF) was realized with the use of the wavelength-dispersive spectrometer (WD-XRF) Axios mAX 4 kW equipped with an Rh source (PANalytical, Malver, UK). The semiquantitative analysis of the XRF spectrum of initial ferrierite materials was carried out with the application of the PANalytical standardless analysis package Omnian. The infrared spectra of ferrierite before and after ion-exchange modification with cobalt cations were collected in the mid-infrared region (MIR) of 4000 to 400 cm^{-1}, using 256 scans at 4 cm^{-1} resolution with the employment of the Vertex 70v FT-IR spectrometer (Bruker, USA). The standard KBr pellet method was applied for sample preparation. The multipoint nitrogen adsorption/desorption measurements at 77 K were performed using the ASAP 2010 Instrument (Micrometric Instruments, USA) with the samples degassing prior to measurement (378 K, 24 h). It was applied in order to establish the textural parameters of zeolite and mesoporous carbon. The Brunauer–Emmett–Teller (BET) and the Barret-Joyner-Halenda (BJH) methods were used in order to establish the specific surface area (SSA) and porosity of particular modifiers, respectively. The analyzer Zetasizer Nano-ZS (Malvern Instruments, UK) with the dynamic light scattering (DLS) technique was employed for ferrierite and mesoporous carbon nanoparticle sizing. The particle size distribution (PSD) parameters were obtained using Mastersizer 2000 ver. 5.60 software. The energy-dispersive X-ray spectroscopy (EDS) attachment of the scanning electron microscope Tescan Vega3 (ThermoFisher Scientific, USA) was applied for elemental analysis of zeolite before and after modification with Co^{2+} ions, as well as observation of the morphology of the zeolite, mesoporous carbon, and modifying layer of the fabricated sensor.

Electrochemical measurement system.—The M161 multipurpose electrochemical analyzer connected with the M164D electrode stand (both mtm-anko, Poland) was implemented in all voltammetric measurements. The dedicated software EALab 2.1 was utilized for the real-time data acquisition. A standard three-electrode voltammetric cell with the volume of 10 mL was composed of the bare or modified GCE (MF-2012, φ = 3 mm, BASi, USA) working electrode, a double-junction silver-silver chloride (Ag | AgCl | 3 M KCl) reference electrode (MINERAL, Poland), and a platinum wire auxiliary electrode. The electrochemical impedance spectroscopy (EIS) measurements were carried out with the use of the μAUTOLAB III analyzer (EcoChemie, the Netherlands). For the recording of impedance spectra, the dedicated NOVA 2.0 software was applied. Since the measurements were carried out in the range of cathodic potentials, the testing solutions were deoxygenated by purging them with argon (99.999%) for 5 min. Between measurements, as well as during the accumulation period, all solutions were stirred (approx. 200 rpm) using a magnetic bar coated with Teflon® placed at the bottom of the cell. The SevenCompact S210 laboratory pH-meter (Mettler Toledo, Switzerland) was employed to adjust the pH value of buffer solutions.

Procedures.— *Zeolite functionalization and the modified electrode fabrication.*—The targeted functionalization of zeolite to impart electrocatalytic activity was based on the transformation of synthetic NH_4-ferrierite to its cobalt form (CoFER) through the ion-exchange process, following previously established procedures.[47] To maximize the replacement of the original NH_4^+ ions in the ferrierite

structure by Co^{2+} cations, 0.5 g of the zeolite sample was connected with 5 ml of 0.1 mol l^{-1} cobalt(II) nitrate aqueous solution. The obtained suspension was shaken for 24 h at 25 °C and centrifuged at 13 500 rpm for 10 min, after which the ion-exchange procedure was replicated. After that, the ferrierite sample was washed several times with 5 ml of double-distilled water and left at room temperature for 24 h to dry. The confirmation of the cobalt(II) ions presence in the ferrierite structure was obtained based on SEM/EDS measurement.

The received CoFER was subsequently used in the preparation of the ZME, with mesoporous carbon (MC) serving as an electron-conducting component of the modifying layer. A 20 mg ml^{-1} modifying suspension of CoFER and MC was prepared in the following steps: (1) mixing of Co-ferrierite and mesoporous carbon in a 1:1 weight ratio, followed by their homogenization via grinding in agate mortar; (2) dissolving of 5 mg of polystyrene in 400 μl of acetone and 600 μl of dichloromethane; (3) weighing 10 mg of the CoFER/MC mixture, its quantitative transfer to the Eppendorf tube, and adding 500 μl of the above-mentioned solvents containing polystyrene; (4) ultrasonic homogenization of the resulting suspension (Ultrasonic bath IS-1, InterSonic, Poland) for 15 min. Meanwhile, the surface of the GCE was prepared for the application of the modification layer by polishing with 0.3 μm alumina powder (Buehler Micropolish II, USA), washing in water, and rinsing several times in acetone. The Co-ferrierite/mesoporous carbon modified glassy carbon electrode (CoFER/MC-GCE) was obtained by careful depositions of 10 μl of the prepared suspension onto the GCE surface. Subsequently, it was left for complete solvent evaporation at room temperature for 24 h. In the same way, the modifying layer containing only Co-ferrierite or mesoporous carbon (10 mg in each case) was prepared and applied on the surface of GCE, resulting in Co-ferrierite modified GCE (CoFER-GCE) and mesoporous carbon modified GCE (MC-CGE), respectively. All electrodes were stored under laboratory conditions between measurements and were ready to work without any mechanical or electrochemical activation.

Voltammetric and impedance measurements.—The electrochemical properties of the fabricated electrodes were specified using CV and electrochemical impedance spectroscopy (EIS) measurements performed in a 0.1 mol l^{-1} KCl solution containing a 0.5 mmol l^{-1} $[Fe(CN)_6]^{4-}$ reversible redox system. The CV voltammograms were recorded in the potential range of −0.3 to 0.85 V with a scan rate (v) varying from 0.006 to 0.5 V s^{-1}, based on which the active surface area of CoFER/MC-GCE, CoFER-GCE, and MC-GCE was calculated. In the EIS measurement, the sinusoidal signals of the frequency ranging from 100 kHz to 25 mHz and 10 mV amplitude, superimposed on the formal potential of the $[Fe(CN)_6]^{3-/4-}$ redox couple, were applied. Key figures of merit were established by fitting the equivalent electrical circuit (EEC) to each recorded impedance spectrum.

In the calibration of the method, the determination of FUR in real samples, as well as in the interference studies, the DPV technique was applied. Prior to the measurements, the pH of the supporting electrolyte was selected, and the instrumental parameters of the DP technique were optimized through univariate analysis. These parameters included the step potential (E_s), pulse amplitude (dE), waiting time (t_w), current sampling time (t_s), and accumulation time (t_{acc}). Unless otherwise stated, DP voltammograms were recorded in the 0.1 mol l^{-1} phosphate buffer solution with pH 6.8 in the potential range from −0.15 to −0.65 V, using the optimized settings: E_s = 5 mV, dE = 40 mV, t_w = 10 ms, and t_s = 10 ms. For each measurement, at least three consecutive scans, realized in the cathodic and subsequently anodic direction, were registered. All voltammetric and impedance measurements were performed in the deoxygenized, tested solution with a total volume of 5 mL.

Samples preparation and analysis.—The OTC furazidin drugs were obtained from a local pharmacy. According to the manufacturer declaration, *Furaginum Hasco* (PPF Hasco-lek S.A., Poland) and *NeoFuragina* (Teva Pharmaceutical Polska, Poland) contained 50 mg of furazidin per tablet, while each tablet of *UroFuragimum Max* (Adamed Consumer Healthcare S.A., Poland) and *NeoFuraginum Max* (Teva Pharmaceutical Polska, Poland) contained 100 mg of FUR. Their preparation for the DPV analysis was carried out by employing the subsequent steps: five tablets of each medicine were weighed and pulverized using an agate mortar; an appropriate quantity of the powder, equivalent to 10 mg of FUR, was transferred to the 100 mL volumetric flask made of amber glass and dissolved in double-distilled water; the resulting solution was homogenized ultrasonically for 15 min, filtered on the syringe filter with a mixed cellulose ester membrane (MCE), having a pore size of 0.22 μm (Alchem Grupa Sp. z o.o., Poland), and diluted tenfold just before conducting the voltammetric measurements.

The certified reference materials of environmental water and biological fluids were prepared for the spiked recovery studies. For this purpose, the CRMs of wastewater and human urine were first filtered using a syringe filter with an MCE membrane and a pore size of 0.22 μm and then spiked with an appropriate volume of 100 mg l^{-1} FUR standard solution to achieve a concentration of 1 mg l^{-1} and 5 mg l^{-1}, respectively. DPV measurements were performed by adding an appropriate volume of spiked samples to the supporting electrolyte (0.1 mol l^{-1} phosphate buffer solution, pH 6.8) to achieve the final concentrations of FUR of 50 μg l^{-1} and 10 μg l^{-1} in the voltammetric cell.

For the synthetic human plasma, which contains a high level of proteins, after its spiking with a known amount of standard FUR solution to obtain a concentration of 2.5 mg l^{-1}, the step of protein precipitation was carried out.[48] After two hours, 800 μl of the enriched plasma sample was mixed with 200 μl of 10% trichloroacetic acid (TCA) to obtain the final concentration of 2%. The mixture was then vortexed for 5 min and centrifuged for 30 min with an approximate speed of 14 500 rpm (Eppendorf, Germany). The resulting supernatant was filtered using the MCE membrane syringe filter (φ = 0.22 μm) immediately before measurement. For voltammetric analysis, the resulting spiked filtrate was transferred to the supporting electrolyte, resulting in the final FUR concentrations of 50 μg l^{-1} and 10 μg l^{-1} in the quartz cell.

The quantification of FUR in all real samples was performed using the standard addition method. This involved the recording of DP voltammograms for the supporting electrolyte (blank), after adding the appropriate volume of diluted pharmaceutical solution or spiked CRMs, and after three successive additions of FUR standard solution (10 mg l^{-1}). At each step, three subsequent scans were registered, averaged, and corrected for background. Based on the standard addition plot, the FUR content in the tested medicaments, as well as the concentration of FUR in spiked CRM samples, were calculated, factoring in the dilution applied.

Results and Discussion

Characterization of electrode modifiers.—According to the most common classification of zeolites, based on rings of SiO_4 and AlO_4 tetrahedra, which are so-called secondary building units (SBU), ferrierite belongs to the group of zeolites in which SBU is presented in the form of 5–1 complex (T_8O_{16} units).[49] The chemical composition of the examined ferrierite, established based on the XRF measurement, was as follows (in wt%): SiO_2–91.5, Al_2O_3–8.1, Na_2O – 0.04, K_2O–0.01, MgO–0.2, CaO–0.02, TiO_2–0.03, Fe_2O_3–0.06, and other – 0.04. The calculated silica to alumina (Si/Al) molar ratio of 19.3 allows ferrierite to be included in the group of high-silica zeolites that possess high thermal and chemical stability, as well as surface selectivity defined as organophilic.[50]

The presence of 5−1 SBU in the structure of ferrierite causes the appearance of the characteristic series of bands in the IR spectra (Fig. 1A). According to the type of vibration realized in the zeolite structure, they are connected with internal Si−O(Si) and Si−O(Al) vibrations in tetrahedra of SiO_4 and AlO_4 or alumino- and silico-oxygen bridges (bands in the region 1200–400 cm^{-1}), as well as with vibrations of overtetrahedral structural units, i.e., rings built

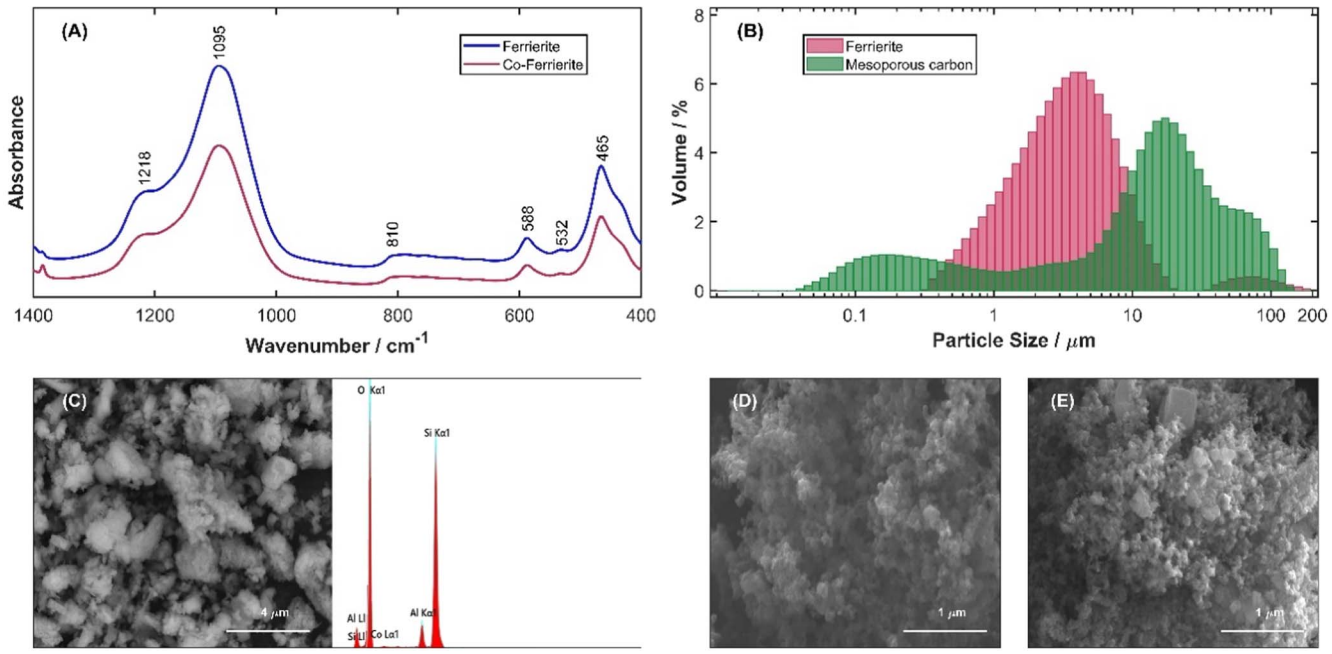


Figure 1. (A) FT-IR spectra of ferrierite before and after ion-exchange modification with cobalt cations in the range of 1400–400 cm^{-1}. (B) Particle size distribution of the employed ferrierite and mesoporous carbon. SEM images of (C) CoFER with corresponding EDS spectrum, (D) MC, and (E) CoFER/MC modifying layer on the surface of GCE.

with a different number of members in the pseudo-lattice range (800–500 cm^{-1}).[51] Based on them, the band located at 588 cm^{-1} can be associated with the characteristic vibration of 5-membered rings in the structure of ferrierite. Comparison of the spectra of ferrierite before and after the ion-exchange process with cobalt cations presented in Fig. 1A shows no noticeable differences between them. The introduction of cobalt ions into the ferrierite structure resulted in small differences in the intensity of the analyzed bands, whereas their position remained unchanged. These observations correspond to the literature data, which indicated that the ion-exchange process alters, first of all, the degree of ring deformation, leading to changes in band intensity in the pseudo-lattice region.[52]

The pore structure of ferrierite is characterized by a two-dimensional (2D) pore system consisting of 10-membered ring (4.2 × 5.4 Å) channels parallel to the [001] direction intersected by 8-membered ring (3.5 × 4.8 Å) channels along the [010] direction. The average pore diameter of the analyzed ferrierite, determined from nitrogen absorption isotherms, equals to 2.56 nm, confirming its medium pore-size structure. Furthermore, ferrierite can be characterized by the micropore volume (V_{micro}) and BET surface area equal to 0.150 $cm^3\ g^{-1}$ and 338.14 $m^2\ g^{-1}$, respectively. On the contrary, mesoporous carbon possesses a much larger average pore diameter (8.87 nm) and pore volume (0.45 $cm^3\ g^{-1}$).

As shown by the particle size distribution (PSD) obtained using the DLS method (Fig. 1B), both modifiers possess the microsized character of their particles. For ferrierite, the particles with a diameter of about 4 μm represent the most abundant size fraction, next to which a minor contribution of the particles with diameters ranging from 30 to 200 μm can be distinguished. On the contrary, mesoporous carbon demonstrated a distinct trimodal PSD. The first size fraction varies from 0.1 to 1 μm with the predominant particles measuring 0.2 μm in diameter, the second from 4 to 40 μm with maximum particle volume of 15 μm of diameter, and the last one fraction from 40 to 150 μm with the highest participation of 60 μm of particle diameter.

The morphological analysis of Co-ferrierite, as well as the corresponding EDS spectrum (Fig. 1C), revealed a plate-like habit of its particles, which exist in the form of irregular aggregates, and confirmed incorporation of Co(II) cations into the zeolite structure. Based on the presence of an additional peak related to Co, the atomic percentage of cobalt content, expressed as the average value from several different surface points, was established at 0.10%. Simultaneously, in the SEM observation of MC (Fig. 1D), the structure with an internal void space and interconnected, spherical particles forming irregular aggregates can be distinguished. Finally, as can be seen in Fig. 1E, the modifying layer containing both CoFER and MC ensured full coverage of the GCE surface, forming rough aggregates of modifiers, resulting in an increase of the surface of the CoFER/MC-GCE and enhancement of its electrochemical activity.

Electrochemical features of fabricated sensors.—To assess the influence of particular components of the modifying layer on the electrochemical performance of the CoFER-GCE, MC-GCE, and CoFER/MC-GCE, the measurements of CV and impedance spectra were performed in 0.1 mol l^{-1} KCl solution in the presence of 0.5 mmol l^{-1} $[Fe(CN)_6]^{4-}$ model redox probe (Figs. 2A and 2B, respectively). By recording voltammograms at various scan rates, the electrochemically active surface area (A_{el}) of the particular electrodes was determined using the Randles-Ševčík equation, assuming a diffusion coefficient of the oxidized and reduced forms (D_O and D_r, respectively) of 7.2 × 10^{-6} $cm^2\ s^{-1}$ for the $[Fe(CN)_6]^{3-/4-}$ redox couple.[53] Additionally, by fitting the equivalent electrical circuit model of a typical Randles cell (shown in the inset of Fig. 2B) to each recorded EIS spectrum, presented as a Nyquist plot, the value of charge transfer resistance (R_{ct}), Warburg impedance module (σ), the differential capacity of the electrical double layer (C_{dl}), and the heterogeneous rate constant of the electrode reaction (k_s) were calculated. The value of R_{ct} and the σ for the impedance spectra measured at the formal potential of the used redox probe can be established using formulas (1) and (2) expressed as [42]:

$$R_{ct} = \frac{RT}{n^2F^2Ak_s}\cdot\frac{2\cdot(D_OD_R)^{1/4}}{D_O^{1/2}\cdot C_O^0 + D_R^{1/2}\cdot C_R^0} \quad [1]$$

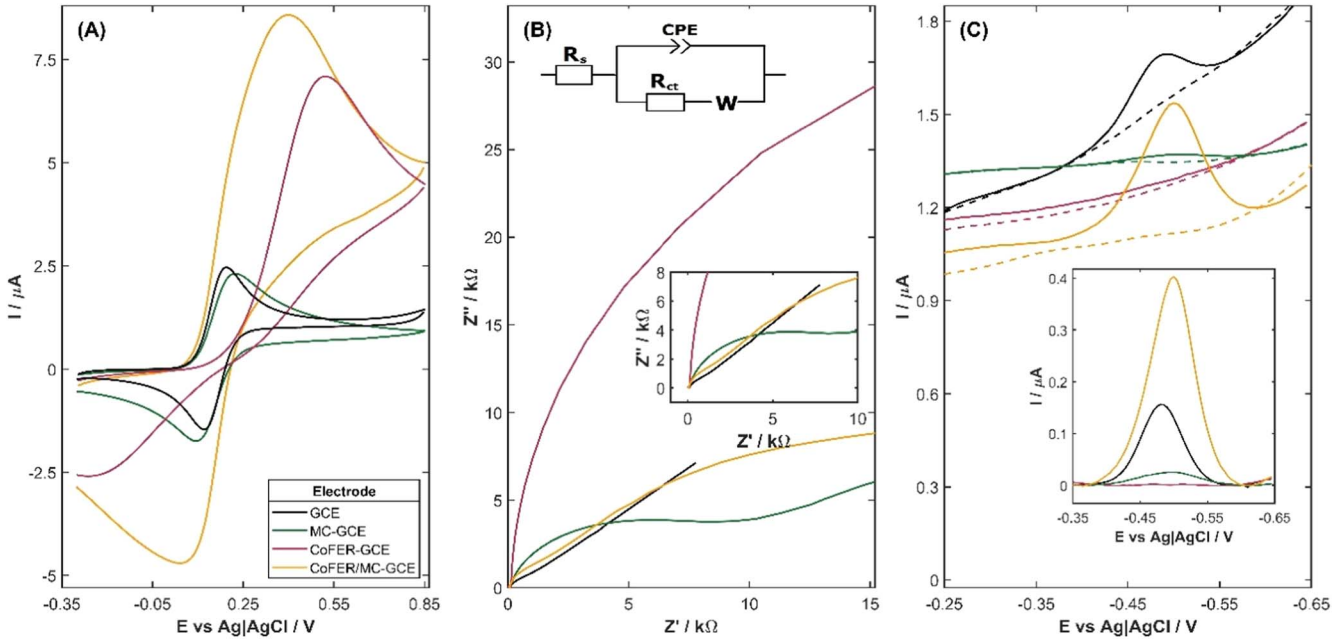


Figure 2. (A) CV curves recorded for 0.5 mmol l^{-1} $K_4[Fe(CN)_6]$ in 0.1 mol l^{-1} KCl solution on the bare GCE, GCE modified with Co-exchanged ferrierite (CoFER-GCE), mesoporous carbon (MC-GCE), and CoFER/MC nanocomposite (CoFER/MC-GCE). Scan rate $v = 20$ mV s^{-1}. (B) Corresponding impedance spectra (Nyquist plot). Inset: Randles cell equivalent circuit. (C) Comparison of DP voltammograms recorded for 0.1 mg l^{-1} FUR ($3.8{\cdot}10^{-7}$ mol l^{-1}) on the GCE, CoFER-GCE, MC-GCE, and CoFER/MC-GCE. Dashed lines represent the background signal. Inset: DP voltammograms subjected to the background correction. Experimental conditions: $E_p = -0.25$ V, $E_k = -0.65$ V, $E_s = 5$ mV, $dE = 40$ mV, $t_w = 10$ ms, $t_s = 10$ ms. Supporting electrolyte: 0.1 mol l^{-1} phosphate buffer solution (pH 6.8).

$$\sigma = \frac{RT}{n^2F^2A\sqrt{2}} \cdot \frac{4}{D_O^{1/2} \cdot C_O^0 + D_R^{1/2} \cdot C_R^0} \quad [2]$$

where R is the universal gas constant (R = 8.314 J mol^{-1} K^{-1}), T is the temperature [K], n is the number of the exchanged electrons per molecule [-], F is the Faraday constant (F = 96485.3 C mol^{-1}), A is the electrode active surface area, D_O and D_R are the diffusion coefficients of the oxidized and reduced forms, respectively, and C_O and C_R are their concentrations in the solution.

Through the division of the Eqs. 1 and 2, it becomes possible to derive the formula for the heterogeneous rate constant k_s (3):

$$k_s = \frac{\sigma}{R_{ct}} \cdot \sqrt{\frac{D_{O/R}}{2}} \quad [3]$$

where $D_{O/R}$ denotes the average diffusion coefficient of the oxidized and reduced forms. The characteristic parameters determined on the basis of the above-mentioned equations are summarized in Table I.

As can be seen in Fig. 2A, the bare GCE and MC-GCE showed a well-defined voltammogram with symmetric anodic and cathodic peaks, for which the value of peak separation (ΔE) is equal to 73 and 92 mV, respectively (Table I). In the case of GCE modified with the layer containing only Co-exchanged ferrierite (CoFER-GCE), an increase in peak current with accompanying very poor symmetry of reduction and oxidation peaks was observed in the recorded cyclic voltammogram. Simultaneously, the very high values of ΔE (0.351 V) and a high ratio of anodic to cathodic peak current ($I_{pa}/I_{pc} = 3.7$) indicate the irreversible character of the $[Fe(CN)_6]^{3-/4-}$ redox reaction on this electrode. These observations are also confirmed in the EIS spectrum of CoFER-GCE, based on which a very high value of R_{ct} and a decrease in the rate constant k_s were observed, proving the resistive character of the zeolite-based modifying layer, resulting from the insulating nature of zeolites. Finally, the introduction of the Co-exchanged ferrierite zeolite to the modifying layer containing MC (CoFER/MC-GCE) resulted in more symmetrical anodic and cathodic peaks with I_{pa}/I_{pc} closer to 1, thus proving the advantage of using zeolite in electrode modification. However, the presence of both CoFER and MC on the surface of GCE leads to an increase in background current, as confirmed by the high value of electrical double layer capacitance $C_{dl,}$ which equals to ca. 70 μF cm^2. This phenomenon is directly connected with the very good sorption properties and ion exchange capacity of zeolites. Among all modified electrodes, CoFER-MC-GCE is distinguished

Table I. The comparison of electrochemical properties of tested electrodes achieved from CV and EIS measurements.

Parameter	Unit	GCE	MC-GCE	CoFER-GCE	CoFER/MC-GCE
Formal potential, E_p	mV	156	159	261	212
Peak separation, ΔE	mV	73	92	351	256
Anodic peak current, I_{pa}	μA	2.4	2.3	6.8	8.3
Anodic to cathodic peak current, I_{pa}/I_{pc}	-	1.0	0.8	3.7	1.4
Charge-transfer resistance, R_{ct}	kΩ	0.8	6.4	69.5	1.7
Warburg coefficient, σ	$k\Omega s^{-1/2}$	8.2	15.1	2.0	12.0
Electrode active surface, A_{el}	mm^2	6.8	3.7	11	15
Double-layer capacitance, C_{dl}	μF cm^{-2}	30.5	5.7	1.9	69.5
Rate constant, k_s	m s^{-1}	1.9 10^{-4}	$4.5{\cdot}10^{-5}$	$5.5{\cdot}10^{-7}$	$1.3{\cdot}10^{-4}$

by one of the lowest values of R_{ct}, and the highest value of the reaction rate constant k_s, which is undoubtedly an advantageous effect, confirming the crucial role of both modifiers in the performance of the presented sensor. The favorable electrochemical properties of CoFER/MC-GCE are also indicated by the expansion of the electrode active surface, which is ca. 2-times higher with respect to the unmodified GCE. Based on these observations, it can be concluded that the improvement of electrical characteristics of CoFER/MC-GCE resulted from the synergistic effect of the Co-exchanged ferrierite and mesoporous carbon.

The comparison of the DPV reduction signal of 0.1 mg l^{-1} FUR in the 0.1 mol l^{-1} phosphate buffer solution with pH 6.8 recorded on the bare GCE, GCE modified with separate components of the modifying nanocomposite, as well as the CoFER/MC-GCE, further confirmed the favorable electrochemical performance of the developed sensor. As presented in Fig. 2C, the unmodified GCE is characterized by the relatively well-developed FUR reduction peak at the potential of −0.482 V vs Ag | AgCl | 3 M KCl reference electrode. Simultaneously, the high background current value results in an unfavorable signal-to-noise (S/N) ratio, consequently leading to low measurement precision and sensitivity. On the contrary, the CoFER-GCE showed no detectable analytical peak of FUR, probably due to the low electrical conductivity of the zeolite. Meanwhile, the mesoporous carbon on its own exhibits negligible electrochemical sensitivity to FUR, which is likely attributed to its extensive dispersion within the matrix. The MC-GCE provided a weak reduction signal of FUR, which became more apparent after the background subtraction step (inset in Fig. 2C). An adequate electrical conductivity and sensitivity of the modifying layer can only be obtained by the combination of mesoporous carbon and zeolite, which act synergistically by the formation of "long" conglomerates of nanoparticles, thus significantly lowering the charge transfer resistance. Consequently of the GCE modification with Co-ferrierite and mesoporous carbon nanocomposite, a significant enhancement of the furazidin reduction peak current was observed. With respect to the unmodified GCE, the CoFER/MC-GCE demonstrates a 2.6-times higher electrochemical response, justifying its modification.

In the next step, the influence of the thickness of the modifying layer on the sensitivity and stability of the sensors was established by the optimization of the volume of modifying suspension deposited on the GCE surface. For this purpose, the CoFER/MC-GCEs were prepared according to Zeolite functionalization and the modified electrode fabrication section. with the following volume of CoFER/MC suspension: 2.5, 5.0, 7.5, 10.0, and 12.5 µl, and subsequently the reduction signal of 0.4 mg l^{-1} FUR was recorded on each electrode. The obtained results (see Supplementary Material, Fig. S1) clearly indicate that the FUR peak current increased with increasing the modifier volume up to 10.0 µl, for which the recorded signal is about 1.3-times higher than for the volume of 2.5 µl. Since further growth in the volume of the deposited modifying layer up to 12.5 µl resulted in an insignificant increase in the signal height, the GCE modified with the CoFER/MC modifying suspension in the volume of 10.0 µl was selected for the next measurements.

Electrochemical behavior of furazidin on the CoFER/MC-GCE.—To investigate the electrochemical behavior of furazidin on the CoFER/MC-GCE, the CV and DPV measurements in 0.1 mol l^{-1} phosphate buffer solution were carried out using various scan rate values (v) and different pH levels of the supporting electrolyte, respectively. Cyclic voltammograms were recorded in the presence of 2 mg l^{-1} of FUR in the potential range of −0.25 to −0.65 V with the scan rate changing from 0.006 to 0.5 V s^{-1}.

As can be seen from the cyclic voltammograms (Fig. 3A), FUR undergoes a reduction process on the CoFER/MC-GCE, displaying a single cathodic peak for each scan rate. The absence of any oxidation peak in the reverse sweep clearly indicates the irreversibility of the studied electrode reaction. To suggest a possible FUR reduction mechanism, a systematic analysis of the impact of v on the recorded cathodic peak current (I_p) and potential (E_p) was carried out. According to Fig. 3A, with an increase in the scan rate value, a higher FUR peak current was observed. By plotting the values of ln(I_p) vs ln(v) (Fig. 3B), the linear dependence with an excellent correlation coefficient (R) equal to 0.9994 was obtained. Its slope was equal to 0.85 ± 0.01, which is close to the theoretical value of 1.0, corresponding to the adsorption-controlled processes.[54]

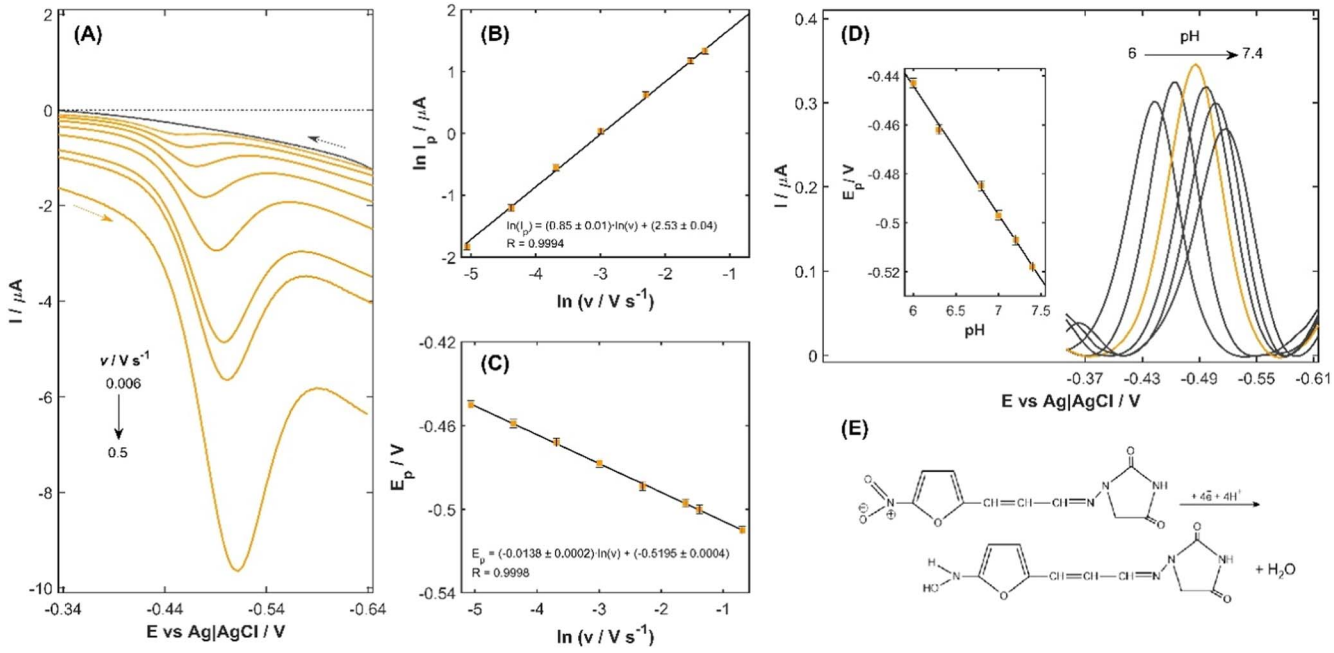


Figure 3. (A) CV curves recorded on CoFER/MC-GCE for the increasing scan rate values in the range of 0.006 to 0.5 V s^{-1} in the presence of 2 mg l^{-1} FUR in 0.1 mol l^{-1} phosphate buffer solution (pH 6.8). Orange lines represent the cathodic scan, whereas the grey one—the anodic scan. The relationship between (B) the natural logarithm of the reduction peak current (I_p) and the natural logarithm of the scan rate (v), and (C) the reduction peak potential (E_p) and the natural logarithm of (v). (D) DPVs registered for the reduction of 0.4 mg l^{-1} FUR in 0.1 mol l^{-1} phosphate buffer solution of pH ranging from 6.0 to 7.4. Inset: The dependence between the peak potential (E_p) and pH of the supporting electrolyte. (E) Proposed mechanism of FUR electrode reaction.

In addition to noted changes in the peak current, the shift in the cathodic peak potential toward more negative values was observed as the scan rate increased. Based on the collected data, the linear dependence between the E_p and the natural logarithm of v with the slope equal to -0.0138 ± 0.0002 was obtained (Fig. 3C). For an irreversible, adsorption-controlled process, the above relationship can be expressed as follows (4)[55]:

$$E_p = K - \frac{RT}{\alpha n F} \ln(v) \quad [4]$$

where K - constant value, α - charge transfer coefficient [-], n - number of electrons [-], ν - scan rate [V s^{-1}].

Assuming physiochemical constants, $T = 298$ K (25 °C), and the obtained value of the slope of E_p vs ln v dependence, the $n\alpha$ value calculated according to Eq. 4 was equal to 1.86. Taking into account the typical adopted value of $\alpha = 0.5$,[54,55] the number of electrons involved in the electrode reaction was estimated to be 3.72, which indicates that four electrons are involved in the reduction of furazidin on the surface of the CoFER/MC-GCE.

In addition, to obtain a comprehensive understanding of the mechanism of the FUR electrode reaction, further measurements were carried out. This involved examining the effect of the pH value of the supporting electrolyte on the peak potential by recording DPV curves in 0.1 mol l^{-1} phosphate buffer solution with pH levels ranging from 6.0 to 7.4 in the presence of 0.4 mg l^{-1} of FUR in the voltammetric cell. The resulting voltammograms, after baseline subtraction, are presented in Fig. 3D. A peak potential undergoes a shift to a more negative value with the increasing pH value of the supporting electrolyte, indicating the participation of protons in the FUR reduction process. The linear relationship established between E_p and pH (inset in Fig. 3D) with the following equation: E_p [V] = $-(0.053 \pm 0.001)$ pH–(0.129 ± 0.008) ($R = 0.9998$) is characterized by a slope value close to the theoretical value of 0.059 V per unit change in the pH, which implies that the number of protons exchanged during the electrode reaction is equal to the number of electrons.

Taking into account all findings, a probable mechanism of furazidin reduction on the surface of CoFER/MC-GCE was proposed and presented in Fig. 3E. Well-defined irreversible reduction peak of FUR is related to the four-electron, four-proton reduction of the nitro group to the hydroxylamine group. This mechanism is in accordance with the well-described process of nitroreduction of the 5-nitrofuran derivatives (i.e., nitrofurazone, nitrofurantoin, furazolidone) to their hydroxylamine counterparts.[56–58]

As can be seen in Fig. 3D, the pH of the supporting electrolyte also affects the FUR peak current. It is clearly visible that in a 0.1 mol l^{-1} phosphate buffer solution with pH in the range of 6.0 to 7.4, the maximum current response of CoFER/MOR-MC was achieved at pH 6.8. Consequently, this pH value was selected as optimal for further voltammetric measurements.

Optimization of the experimental conditions.—To ensure the optimal measurement conditions for the determination of FUR on the CoFER/MC-GCE, a univariate optimization of the instrumental parameters of the DPV technique was performed. The influence of measurement conditions on the reduction signal was evaluated using a supporting electrolyte solution with a FUR concentration of 0.4 mg l^{-1}. The following instrumental parameters were systematically varied: step potential (E_s) from 1 to 6 mV, pulse amplitude (dE) in the range of 10 to 60 mV (positive and negative mode), current sampling time (t_s) from 5 to 20 ms, and waiting time (t_w) in the range of 5−30 ms. After comparing the peak current, shape, position, symmetry, and the signal-to-noise ratio, the optimal parameters were identified as follows: $E_s = 5$ mV, $dE = 40$ mV, $t_s = 10$ ms, and $t_w = 10$ ms, which were then applied in subsequent experiments.

Analytical performance.—In order to provide the best analytical performance of the CoFER/MC-GCE in the determination of furazidin, the preconcentration step was introduced during the measurement. The influence of the preconcentration time (t_{acc}) on the current response of furazidin was examined in the range of 0 to 180 s for three different concentrations of FUR: 0.1; 0.04, and 0.01 mg l^{-1} (Fig. 4A). As can be seen, for each concentration, the lowest peak current of the FUR reduction was registered for the measurement without the preconcentration step (t_{acc} equal to 0 s). Simultaneously, with the extension of t_{acc}, an increase in the value of FUR peak current was observed, indicating an enhancement of FUR accumulation at the CoFER/MC-GCE surface. The preconcentration

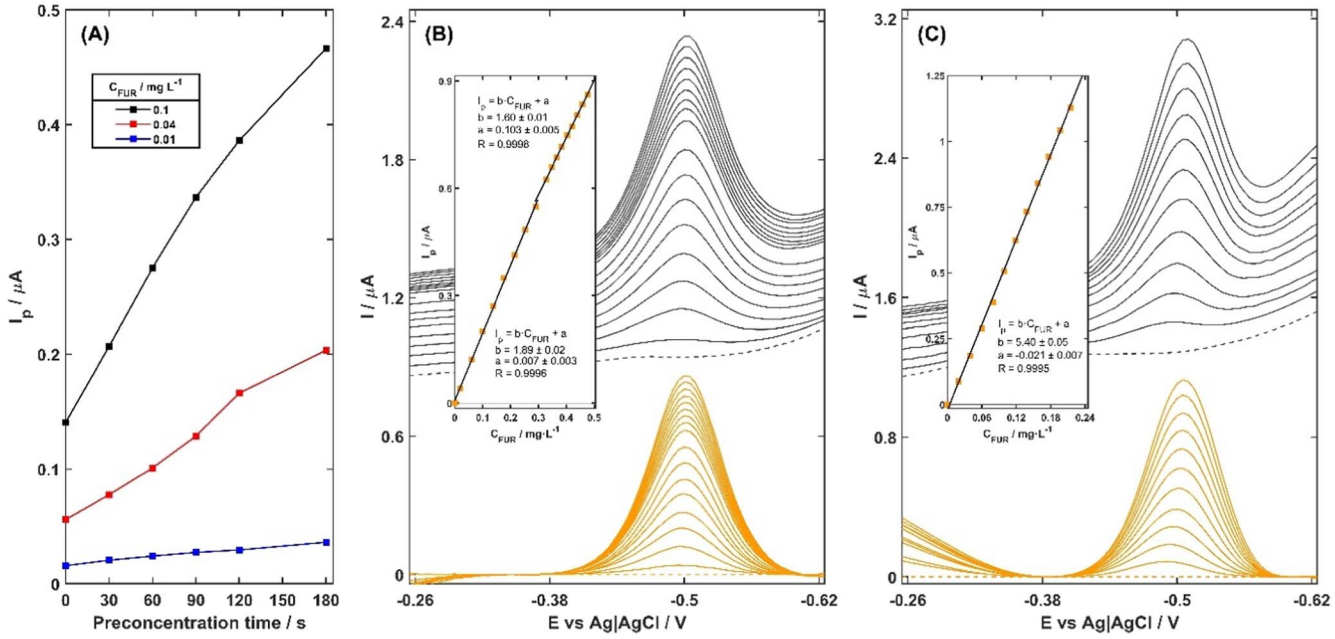


Figure 4. (A) Dependence of furazidin peak current on the preconcentration time in the range from 0 to 180 s for different FUR concentration in 0.1 mol l^{-1} phosphate buffer solution of pH 6.8. DP voltammograms registered on CoFER/MC-GCE for the increasing concentration of FUR in range from blank (dashed line) to (B) 0.48 mg l^{-1}, and (C) 0.215 mg l^{-1} (grey lines) with the curves obtained after the background correction (orange lines). Inset: corresponding calibration plots. Accumulation time $t_{acc} = 0$ s (B) and $t_{acc} = 60$ s (C). The others experimental conditions as in Fig. 2C.

Table II. Analytical parameters obtained for the developed procedures for furazidin determination.

Parameter	Unit	Figure 5B	Figure 5C
		0.02–0.29	
LR	mg l^{-1}	0.33–0.48	0.02–0.22
Intercept *a*	μA	0.007 ± 0.003	−0.021 ± 0.007
		0.103 ± 0.005	
Slope *b*	μA L mg^{-1}	1.89 ± 0.02	5.40 ± 0.05
		1.60 ± 0.01	
R	–	0.9996	0.9995
		0.9998	
LOD	μg l^{-1} (mol l^{-1})	4.8 ($1.8{\cdot}10^{-8}$)	3.9 ($1.5{\cdot}10^{-8}$)
LOQ	μg l^{-1} (mol l^{-1})	15.9 ($6.2{\cdot}10^{-8}$)	13.0 ($4.9{\cdot}10^{-8}$)

LOD = 3 SD/b, and *LOQ = 10 SD/b*, where *SD*: the standard deviation of y-intercepts of the regression line, and *b*: the slope of the calibration curve.

step impacts not only sensitivity and the detection limit, but also determines the practicality of the method. In that respect, a t_{acc} value of 60 s was selected, as it ensures a trade-off between the relatively high sensitivity and reasonably short time of measurements.

The analytical performance of the proposed method was established through a calibration study carried out in a 0.1 mol l^{-1} phosphate buffer solution of pH 6.8 under optimized conditions with and without the application of the preconcentration step. The DP voltammograms for the increasing concentration of FUR obtained on the CoFER/MC-GCE for the t_{acc} of 0 and 60 s are presented in Figs. 4B and 4C, respectively. For each registered curve, the background correction step was applied by subtracting the background signal obtained in the supporting electrolyte without the presence of the depolarizer (dashed line in Fig. 4), resulting in the orange-marked curves. Based on them, the linear dependences between the peak current and the FUR concentration were plotted (inset in Figs. 4B and 4C), and coefficients of the calibration curves, as well as the validation parameters, such as sensitivity, linear range (*LR*), limit of detection (*LOD*), and limit of quantification (*LOQ*), were established and summarized in Table II.

As shown in Fig. 4B, for the measurements performed without the preconcentration step ($t_{acc} = 0$ s), the reduction peak current was proportional to the FUR concentration over two linear ranges: 0.02 – 0.29 mg l^{-1} ($7.2{\cdot}10^{-1}$ – $1.1{\cdot}10^{-1}$ mol l^{-1}) and 0.33 – 0.48 mg l^{-1} ($1.2{\cdot}10^{-1}$ – $1.8{\cdot}10^{-1}$ mol l^{-1}) with a correlation coefficient equal to 0.9996 and 0.9998, respectively. Implementing 60 s of the preconcentration time resulted in a nearly 3-fold increase of the measurement sensitivity and lowering of the detection and quantification limit. Therefore, the improvement of detection capabilities achieved by the accumulation of furazidin on the surface of the CoFER/MC-GCE allows the recognition of small variations of FUR concentration with high precision, which is extremely desirable in the analysis of pharmaceutical, biological, and environmental samples with complex matrices. The *LOD* and *LOQ* estimated to be 3.9 μg l^{-1} ($1.5{\cdot}10^{-1}$ mol l^{-1}) and 13.00 μg l^{-1} ($4.9{\cdot}10^{-8}$ mol l^{-1}), respectively, can be improved by further extension of the accumulation time, creating the possibility of determining nanomolar concentration of FUR.

According to the current literature, the voltammetric determination of furazidin has not been performed to date. Compared to previously reported methods,[16–19] only the amperometric enzyme biosensor allows to obtain the LOD value at the nanomolar level. However, the main drawbacks of this analytical tool are the need of electrode pretreatment to obtain a reproducible surface, as well as fouling of the surface of transducers due to prolonged use, especially in complex biological matrices, which affect the biosensor response.[59] In this regard, the simple drop-casting method of obtaining the proposed CoFER/MC-GCE leads to the achievement of a voltammetric sensor ready to use without any mechanical or electrochemical pretreatment and activation. Additionally, the fabricated sensor can be applied in a considerable number of measurements without any deterioration in properties. For five consecutive DP voltammograms recorded for 80 μg l^{-1} FUR on ten separate copies of this electrode, the value of the relative standard deviation (RSD) of the peak current fell in the range between 0.7% and 1.8%, demonstrating the exceptional repeatability and short-time stability of every prepared electrode. Furthermore, the RSD value of the peak current obtained at all series of ten fabricated electrodes was equal to 7.5%, which confirms the very good reproducibility of CoFER/MC-GCE and its preparation procedure. The proposed sensor can also be characterized by the long-term stability, as shown by the small variation of peak current after its storage for 7 and 14 days under laboratory conditions, which did not exceed 19%.

Selectivity.—The selectivity of the developed protocol was evaluated through interference studies using a series of excipients and an active substance typically present in the matrix of pharmaceutical, biological, and environmental samples. For this purpose, the reduction signal of 80 μg l^{-1} FUR was recorded in the presence of a 10- to 3000-fold excess of particular inorganic ions (K^+, Na^+, Mg^{2+}, Ca^{2+}, Cu^{2+}, Pb^{2+}, Cd^{2+}, Cl^-, NO_3^-, PO_4^{3-}, and SO_4^{2-}) and 15-, 75-, 125-, and 250-fold excess of humic acids (HA), uric acid (UA), and surface-active compounds, i.e., sodium dodecyl sulphate (SDS), cetrimonium bromide (CTAB), and Triton-X100, compared to the concentration of furazidin. In the case of selected interferents representing the basic excipients of drugs (titanium dioxide, starch, magnesium stearate, and cellulose), 5 mg of each powdered substance was added directly to the voltammetric cell. The tolerable limit was defined as the concentration of the interferent causing a deviation in the range of ± 5% of the FUR peak current.

Based on the conducted research, it was found that inorganic cations and anions did not affect the reduction signal of FUR, even for a 3000-fold excess compared to the FUR concentration. The exception was observed in the case of heavy metal cations, in particular for cadmium and lead ions, which caused a clearly visible amplification of the analytical signal to 180% and 160% of the initial value for their 500-fold excess, respectively. Simultaneously, the change in the FUR reduction potential was observed, which shifted from −0.516 to −0.500 V in the absence and presence of heavy metals, respectively. Other findings were made for copper cations, the presence of which resulted in a systematic decrease in FUR reduction peak current, reaching 15% of its initial value for the highest excess.

The suppression effect of the analytical signal was noted in the case of surface-active compounds, uric acid, starch, and cellulose. With the successive addition of Triton-X100 (non-ionic surfactant), SDS (anionic surfactant), and CTAB (cationic surfactant), the recorded peak current decreased, reaching 31%, 38%, and 65%, respectively, of its initial value in the case of the highest concentration of these interferents. Furthermore, in the case of CTAB, the shift of the FUR peak potential from −0.515 to −0.473 V was observed. The similar suppressing effect of FUR reduction signal was found for the addition of UA—for its 250-fold excess compared to FUR concentration (80 μg l^{-1}), the recorded peak current decreased to 52% of its initial value. In turn, in the presence of 5 mg of starch or cellulose in the voltammetric cell, the analytical signal was suppressed to 70% and 58% of its initial value, respectively.

Finally, upon the addition of HA, titanium dioxide, and magnesium stearate, an increase in the height of the analytical signal was observed. The most significant strengthening of the recorded peak current was noted for the 250-fold excess of HA, the presence of which resulted in an increase in the FUR reduction signal by 80%. A slightly less enhancing effect was observed in the case of titanium dioxide and magnesium stearate, for which recorded FUR signal increased to 136% and 117% of its initial value, respectively.

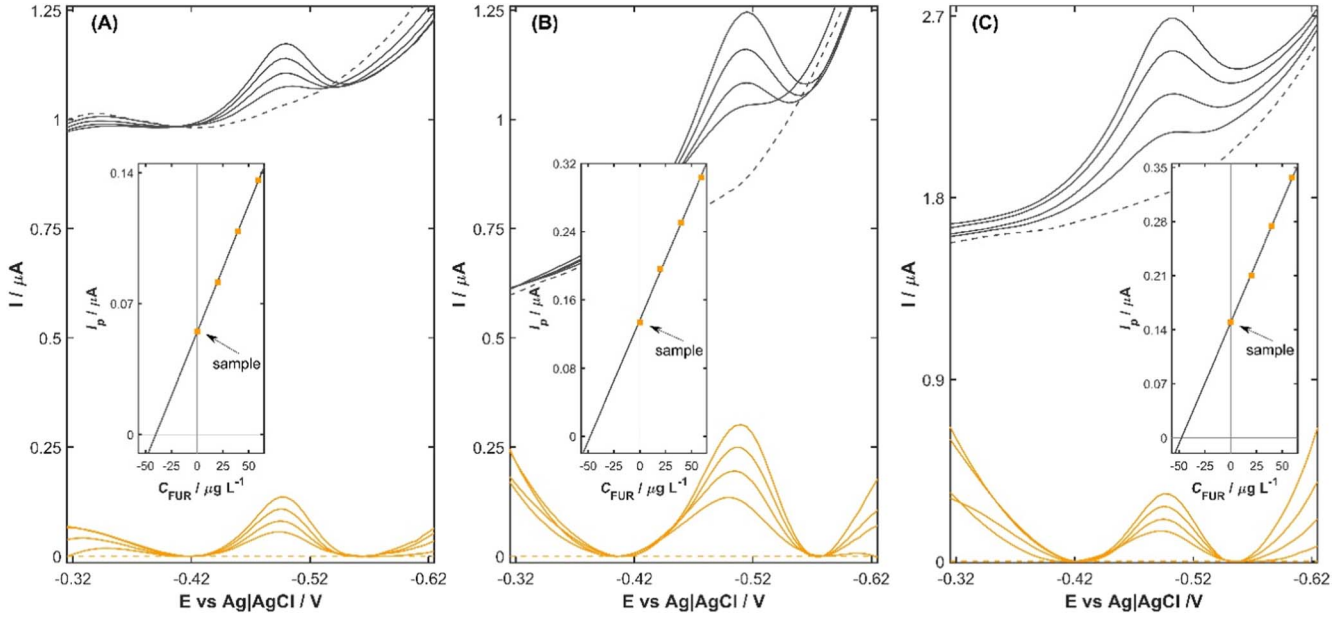


Figure 5. Investigation of the real samples: (A) tablets of *NeoFuragina* 50 mg, CRMs of (B) wastewater, and (C) human urine spiked with 50 μg l^{-1} FUR. The experimental voltammograms were depicted in black, whereas the curves obtained after the background correction are marked in orange. Inset: corresponding calibration plots. Experimental conditions of DPV measurement as in Fig. 2C.

Analysis of the real samples.—To verify the applicability of the devised protocol, the CoFER/MC-GCE was used in the determination of FUR in real samples with pharmaceutical, biological, and environmental matrices (Fig. 5). Each tested sample, after undergoing initial preparation (see Samples preparation and analysis section), was analyzed according to the standard addition method. To verify the accuracy of the analysis, based on the manufacturer's declaration, the relative error (RE) and recovery parameter have been calculated for medications and spiked CRMs of wastewater and biological fluids, respectively. The obtained results are summarized in Table III.

For each type of sample, a single peak was detected within the potential range associated with FUR reduction, (Fig. 5), which indicates that the developed protocol is selective towards determination of the FUR. Furthermore, the observed increase in current with each standard addition demonstrates that matrix components did not block the active center on the surface of the CoFER/MC-GCE. As a results of that, it can be concluded that the proposed voltametric strategy for FUR determination is suitable for the analysis of pharmaceutical, biological, and environmental samples.

The abovementioned statement was confirmed by the obtained FUR content in the analyzed samples (Table III). In the case of pharmaceuticals, very good compliance with the content given by the manufacturer was observed, as shown by the RE value not exceeding 2.3%. Taking into account the excellent precision (RSD $< 2.1\%$), it can be stated that the proposed protocol allows for accurate and reliable determination of FUR in medicaments. The same statement can be made with respect to the CRMs of human serum and urine, and wastewater spiked with a known amount of furazidin. The computed recoveries ranged from 79.0 to 112.0% with RSD values lower than 3%, demonstrating the precision, accuracy, and stability of the devised method of FUR analysis in complex matrices with the use of CoFER/MC-GCE.

Table III. Results of furazidin determination in the pharmaceuticals and CRMs of biological fluids and waste waters.

Sample	Declared value [mg per tablet]	Amount found [mg per tablet]	RE [%]	RSD [%]
Furaginum Hasco	50	50.1 ± 1.0	0.24	2.1
NeoFuragina	50	50.4 ± 0.7	0.84	1.3
UroFuragimum Max	100	97.8 ± 1.3	−2.25	1.3
NeoFuraginum Max	100	100.2 ± 1.0	0.23	0.9

Sample	Concentration of FUR [μg l^{-1}]		Recovery [%]	RSD [%]
	Added	Found x ± s		
Waste Water	50	50.9 ± 0.7	101.8	1.4
SPS-WW1	10	10.8 ± 1.1	108.0	2.6
Waste Water	50	49.5 ± 1.4	99.0	2.8
SPS-WW2	10	9.3 ± 0.9	93.0	3.7
Serum	50	51.5 ± 0.8	102.9	1.5
	10	11.2 ± 2.1	112.0	3.9
Urine	50	49.4 ± 0.7	98.7	1.4
	10	7.9 ± 3.4	79.0	4.8

$\bar{x}$—mean value; s—standard deviation; RSD% $= s/\bar{x} \cdot 100\%$

Conclusion

To date, no voltammetric approaches have been reported for the determination of the antibacterial drug furazidin. Therefore, a new, highly sensitive, and selective DPV protocol using CoFER/MC-GCE as the working electrode was presented for the first time. The developed procedure of GCE modification, based on the simple and low-cost method of drop-casting the suspension of the Co-exchanged ferrierite and mesoporous carbon nanocomposite, allows to achieve a durable layer, offering excellent repeatability, short- and long-term stability, as well as good reproducibility and accuracy of analytical investigations. As shown by the results of textural and EIS measurements, the favorable electrochemical properties of the CoFER/MC-GCE arises from the combination of unique features of zeolite and mesoporous carbon. On the basis of the CV investigation, the electrochemical behavior of FUR on the CoFER/MC-GCE was established as an irreversible four electron—four proton, adsorption-controlled reduction process.

Through optimization of the measurement conditions and pH value of the supporting electrolyte, the proposed DPV protocol for the determination of FUR is distinguished by a high sensitivity and low limit of detection, which equals to 3.9 $\mu g\, l^{-1}$ ($1.5{\cdot}10^{-8}$ mol l^{-1}). The introduction of an accumulation step into the proposed DPV strategy, improving the detection capabilities, enables the determination of trace amounts of FUR in the complex matrices without elaborate sample preparation. The last was confirmed by a successful quantitative analysis of pharmaceuticals, CRMs of human urine, serum, and wastewater with the recovery ranging from 79.0% to 112.0%.

In summary, the proposed for the first time voltammetric assay of furazidin determination in conjunction with the one-step, inexpensive method of CoFER/MC-GCE fabrication offers excellent sensitivity, accuracy, low detection limit, high reproducibility, low cost, and short time of direct detection and determination of FUR in samples of pharmaceutical, biological, and environmental origin.

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